

Project Design Phase

Proposed Solution

Date: 26 June 2026

Team ID: XXXXX

Project Name: Smart Lender

Maximum Marks: 2 Marks

Proposed Solution

S.No.	Parameter	Description
1	Problem Statement (Problem to be Solved)	Banks and financial institutions often rely on manual verification of loan applications, making the approval process slow, time-consuming, and susceptible to human error. Applicants also face long waiting periods and uncertainty regarding their loan eligibility.
2	Idea / Solution Description	Develop a web-based Smart Lender application using Python, Flask, HTML, CSS, and a Random Forest Machine Learning model . The system collects applicant information, processes the data, and predicts whether the loan is likely to be approved or rejected in real time.
3	Novelty / Uniqueness	The solution combines Machine Learning with an easy-to-use web interface to automate loan eligibility prediction. It provides quick, consistent, and data-driven decisions, reducing manual effort while improving prediction accuracy and user experience.
4	Social Impact / Customer Satisfaction	The system benefits both banks and applicants by reducing processing time, improving transparency, minimizing human errors, and delivering faster loan approval decisions. This enhances customer satisfaction and supports fair lending practices.
5	Business Model (Revenue Model)	The Smart Lender application can be offered to banks, financial institutions, and fintech companies as a Software-as-a-Service (SaaS) solution through subscription plans, annual licensing, or customized enterprise deployments with maintenance and technical support.
6	Scalability of the Solution	The application is designed to support increasing numbers of users and larger datasets. Future enhancements can include user authentication, database integration, cloud deployment, document verification, real-time credit score integration, and AI-powered loan recommendation features.